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SciScore Report

Below you will find your SciScore report containing three tables. Your score is calculated based on adherence to scientific rigor criteria (Table 1) and identification of key biological resources (Table 2). Table 3 contains statistical tests and oligonucleotides but is not scored. If SciScore makes any mistakes, please contact us to help us learn and improve.

Table 1: Rigor Adherence Table

Ethics
IACUC: All producers were approved by Institutional Animal Care and Use Committee of the Fifth Affiliated Hospital of Sun Yat-sen University.
Euthanasia Agents: In brief, when rats were anesthetized with 40 mg/kg sodium pentobarbital, the left common artery and the left external carotid artery were exposed.
Inclusion and Exclusion Criteria
not detected.
Attrition
not detected.
Sex as a biological variable
MATERIALS AND METHODS Animal model Adult male SD rats (8~10 weeks, approximately 270±17 g) and 17-day-old SD rats were all obtained from the Experimental Animal Center of the Chinese Academy of Medical Sciences, People’s Republic of China.
Subject Demographics
Weight: MATERIALS AND METHODS Animal model Adult male SD rats (8~10 weeks, approximately 270±17 g) and 17-day-old SD rats were all obtained from the Experimental Animal Center of the Chinese Academy of Medical Sciences, People’s Republic of China.
Age: MATERIALS AND METHODS Animal model Adult male SD rats (8~10 weeks, approximately 270±17 g) and 17-day-old SD rats were all obtained from the Experimental Animal Center of the Chinese Academy of Medical Sciences, People’s Republic of China.
Randomization
A blinded investigator used a microscope to take images and chose images randomly for each section .
Blinding
A blinded investigator used a microscope to take images and chose images randomly for each section .
Power Analysis

not detected.
Replication
Statistical analysis All experiments were performed in triplicate.
Number: Statistical analysis All experiments were performed in triplicate.

Table 2: Key Resources Table

Your Sentences	REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies			
Immunohistochemistry staining The brain tissues were fixed with 4 % formaldehyde and cut into 5 mm-thick serial sections , and immunohistochemistry staining was performed according to the immunohistochemical kit (Boster Biological Technology , Wuhan , China) by using the specific cleaved caspase-3 antibody (1:300; ab2302 , Abcam) .	cleaved caspase-3		
After blocking with 5 % non-fat milk , the membrane was incubated with primary antibody against TXNIP (1:500 , ab188865 , Abcam) and GAPDH (1:1,000 , ab9485 , Abcam) at 4# overnight .	TXNIP		
	GAPDH		
Recombinant DNA			
To determine the interaction between miR-20b and TXNIP , the wild type (WT) and mutant type (MUT) 3'-UTR of TXNIP containing putative binding sites of miR-20b were amplified by PCR and cloned into pmirGLO dual luciferase reporter vector (Promega , Madison , WI , USA) .	pmirGLO dual luciferase		
Software and Algorithms			
Finally , the proteins of interest were detected by using an enhanced chemiluminescence (ECL) kit (Pierce , Rockford , IL , USA) and band intensity was quantified by Image-Pro Plus 6.0 software (Media Cybernetic)	Image-Pro Plus		

Other Entities Detected

Your Sentences	Recognized Entity
Statistical Tests	
Difference between the two groups was analyzed by the unpaired Student's t test, and difference among multiple groups was determined by using one-way ANOVA. p value<0.05 was considered as the significant threshold.	t test
	ANOVA
Oligonucleotides	
The sequences were listed as follows: sh-TXNIP sense: 5'-GATCCGCCAGCCAACTCAAGA GGCAAAGAAATTCAAGAGATTTCT TTGCCTCTTGAGTTGGCTGGTTTT TTG-3' , anti-sense: 5'-AATTCAAAAAACCAGCCAACT CAAGAGGCAAAGAAATCTCTTGAA TTTCTTTGCCTCTTGAGTTGGCTG GCG-3'; sh-NC sense: 5'-GATCCGCCACAACAACTGGAG AAACGCGAAATTCAAGAGATTTCG CGTTTCTCCAGTTGTTGTGGTTTT TTG-3' , anti-sense: 5'-AATTCAAAAAACCACAACAAC TGGAGAAACGCGAAATCTCTTGAA TTTCGCGTTTCTCCAGTTGTTGTG GCG-3' .	5'-GATCCGCCAGCCAACTCAAGA GGCAAAGAAATTCAAGAGATTTCT TTGCCTCTTGAGTTGGCTGGTTTT TTG-3'
	5'-AATTCAAAAAACCAGCCAACT CAAGAGGCAAAGAAATCTCTTGAA TTTCTTTGCCTCTTGAGTTGGCTG GCG-3'
	5'-GATCCGCCACAACAACTGGAG AAACGCGAAATTCAAGAGATTTCG CGTTTCTCCAGTTGTTGTGGTTTT TTG-3'
	5'-AATTCAAAAAACCACAACAAC TGGAGAAACGCGAAATCTCTTGAA TTTCGCGTTTCTCCAGTTGTTGTG GCG-3'
The primers used in this study were listed as follows: miR-20b forward: 5'-TGCAGTAGTTTTGGCATGA-3 ' , reverse: 5'-TCAACAAGAGATTTGTTATCC AAGAG-3'; TXNIP forward: 5'-AGTTACCCGAGTCAAAGCCG-3' , reverse: 5'-TCTCGTTCTCACCTGTAGGC-3'; GAPDH forward: 5'-TGTTGCCATCAATGACCCCTT -3' , reverse: 5'-CTCCACGACGTACTCAGCG-3 '; U6 forward: 5'-GCTTCGGCAGCACATATACT-3' , reverse: 5'-AACGCTTCACGAATTTGCGT- 3' .	5'-TGCAGTAGTTTTGGCATGA-3 '
	5'-TCAACAAGAGATTTGTTATCC AAGAG-3'
	5'-AGTTACCCGAGTCAAAGCCG- 3'
	5'-TCTCGTTCTCACCTGTAGGC- 3'
	5'-TGTTGCCATCAATGACCCCTT -3'
	5'-CTCCACGACGTACTCAGCG-3 '
	5'-GCTTCGGCAGCACATATACT- 3'
	5'-AACGCTTCACGAATTTGCGT- 3'

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